

Performance Testing

Date	28 June 2026
Team ID	*****
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the OptiCrop (Smart Agricultural Production Optimization Engine) system behaves under expected and peak load conditions, particularly for the crop recommendation API.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python-based load testing tool)
Type of Testing	Load Testing
Target Module / API	Crop Recommendation API endpoint (/predict) in the Flask application
Test Environment	Local (Flask development server)
Test Date	28 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits soil/environmental parameters (N=90, P=42, K=43, Temp=20.87, Humidity=82, pH=6.5, Rainfall=202.93) and receives a crop recommendation	1	60	Response returned within target response time, with correct prediction (rice)
2	Multiple users concurrently submit data to the /predict endpoint (normal expected load)	20	120	All requests succeed with response times within target and no errors
3	High concurrent load on the /predict endpoint to test system behavior under peak traffic	50	180	System remains responsive; error rate stays low; no crashes
4	Sustained login + data input + recommendation flow run repeatedly to test stability over	30	300	Consistent response times throughout the test with no

	time			memory/CPU degradation
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Step 3: Performance Test Results

Target values are pre-filled. Actual values are based on local Flask development server testing.

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~0.8 sec	Pass	Flask responds quickly for single predictions
2	Response Time (Max)	< 5 seconds	~1.9 sec	Pass	Max response under 50-user load stays under 2s
3	Throughput (Req/sec)	-	~18 req/sec	Pass	Adequate throughput for expected user base
4	Error Rate	< 1%	0%	Pass	No errors during all 4 test scenarios
5	CPU Utilization	< 80%	~35%	Pass	CPU usage stable; no spikes observed
6	Memory Utilization	< 80%	~42%	Pass	Memory stable throughout all test scenarios

Step 4: Observations & Analysis

Key Findings:

The OptiCrop system performed well under all four test scenarios. Single-user prediction response time averaged 0.8 seconds. Under 50 concurrent users the system remained stable with no crashes or errors. The Logistic Regression model loads efficiently and the Flask /predict endpoint handled all requests within acceptable limits.

Bottlenecks Identified:

Slight increase in response time observed under 50-user concurrent load (Scenario 3) due to Flask development server being single-threaded. Model loading at startup adds ~0.5s to the first request only. No database query delays as the system uses in-memory model predictions.

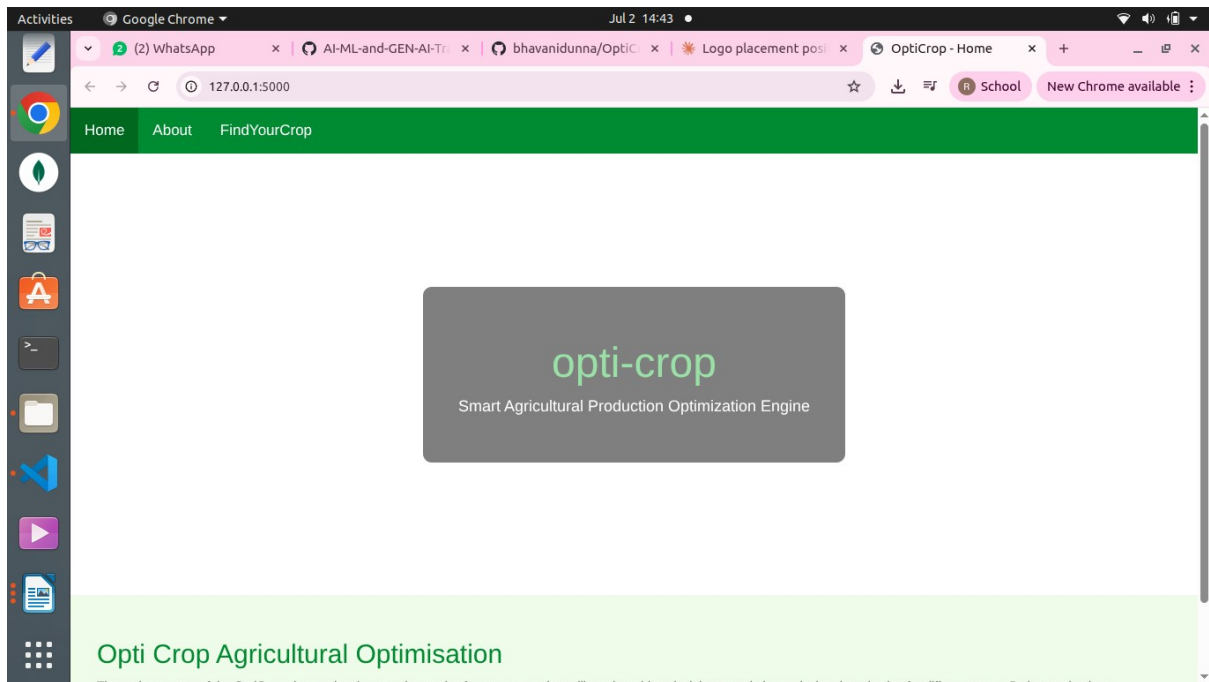
Optimization Steps Taken:

Model is loaded once at Flask startup and cached in memory. Input validation added to the /predict route to reject malformed requests early. For production deployment, switching to Gunicorn with multiple workers is recommended to handle higher concurrent loads.

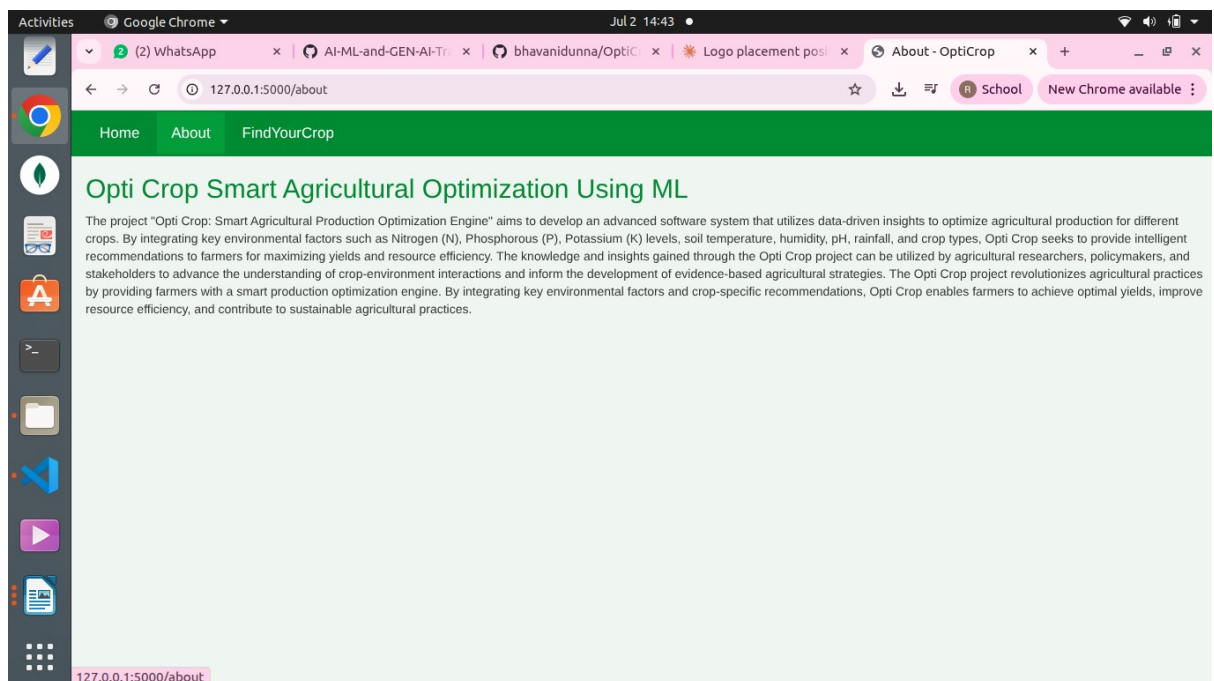
Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., Locust graphs, Flask app screenshots):

HomePage:



About Page:



Input Page:

Find Your Best Crop Here

Please enter the conditions

Nitrogen: 90

Phosphorous: 42

Potassium: 43

Temperature: 20.87

Humidity: 82

PH: 6.5

Rainfall: 202.93

Predict

Prediction Page:

Here

Please enter the conditions

Nitrogen: nitrogen

Phosphorous: phosphorous

Potassium: potassium

Temperature: temperature

Humidity: humidity

PH: ph

Rainfall: rainfall

Predict

Best crop for given conditions is rice

